

BISTRITA, RomaniaWMO: **150850**Lat: **47.1492N**Lon: **24.5139E**Elev: **367**StdP: **96.99**Time Zone: **2.00 (EUE)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.7	-11.6	-17.6	0.8	-12.7	-14.6	1.1	-10.4	9.1	1.3	7.0	-2.7	0.4	50	0.457

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.7	31.2	20.0	29.8	19.7	28.3	19.2	21.6	28.5	20.8	27.6	20.1	26.3	1.7	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.5	14.9	23.8	18.6	14.1	23.1	17.9	13.5	22.4	64.6	28.7	61.7	27.6	59.1	26.3	24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.6	4.9	3.7	-18.9	33.7	3.8	2.7	-21.7	35.6	-23.9	37.1	-26.1	38.7	-28.8	40.6		
(5)			-19.0	23.3	3.8	0.8	-21.8	23.9	-24.0	24.3	-26.1	24.8	-28.9	25.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	-2.1	0.8	4.4	10.4	15.2	18.9	20.1	20.3	15.6	10.0	4.8	-0.9		
	DBStd	8.89	4.48	4.31	4.06	4.43	3.42	3.10	2.87	3.36	4.37	4.05	4.57	4.70		
	HDD10.0	1421	374	259	177	48	3	1	0	0	4	53	164	337		
	HDD18.3	3331	632	492	431	238	107	29	18	19	104	259	407	596		
	CDD10.0	1360	0	0	4	61	164	268	312	319	172	52	8	0		
	CDD18.3	230	0	0	0	1	9	47	71	80	22	0	0	0		
	CDH23.3	2386	0	0	0	21	128	459	673	841	258	6	0	0		
	CDH26.7	718	0	0	0	1	16	116	188	325	73	0	0	0		
(14) Wind	WSAvg	1.2	0.9	1.0	1.5	1.6	1.5	1.3	1.3	1.3	1.2	1.0	0.9	1.0		
(15) Precipitation	PrecAvg	703	45	35	39	54	76	93	88	66	53	49	49	50		
	PrecMax	975	114	93	112	113	217	191	195	191	153	123	110	126		
	PrecMin	418	7	0	2	9	14	12	16	7	2	1	0	0		
	PrecStd	128	25	20	25	24	40	37	41	43	35	33	25	28		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.9	14.5	19.6	25.6	28.5	31.3	32.2	34.3	31.3	24.1	19.1	10.0		
		MCWB	5.6	8.8	10.9	14.5	17.8	21.1	20.9	20.7	18.3	15.5	12.7	7.5		
	2%	DB	6.5	11.0	16.2	23.4	26.5	29.7	30.4	32.0	28.8	22.4	16.2	7.5		
		MCWB	4.5	7.2	9.5	13.9	17.4	20.4	20.3	19.5	17.7	15.0	11.4	6.0		
	5%	DB	4.7	8.5	13.2	20.8	24.6	27.9	28.8	30.2	26.4	19.8	13.7	5.7		
		MCWB	3.3	5.9	7.8	12.9	16.7	19.7	19.7	19.6	17.0	14.2	10.3	4.8		
	10%	DB	3.0	6.6	10.9	18.5	22.7	26.0	27.2	28.3	23.8	17.0	11.4	4.4		
		MCWB	2.1	4.6	6.7	12.0	15.9	19.0	19.6	19.2	16.7	13.1	9.1	3.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.6	9.1	11.9	15.8	19.7	22.7	22.4	22.5	20.2	17.4	13.2	7.7		
		MCDB	7.9	13.4	18.6	22.0	25.2	29.6	28.5	30.3	28.6	20.9	16.8	9.0		
	2%	WB	4.8	7.6	9.9	14.6	18.4	21.1	21.5	21.3	18.9	15.8	11.9	6.2		
		MCDB	6.3	10.4	14.9	21.6	24.2	27.8	28.0	29.1	25.6	20.6	15.0	7.2		
	5%	WB	3.4	6.2	8.6	13.5	17.4	20.3	20.8	20.5	17.9	14.6	10.7	4.9		
		MCDB	4.5	8.3	12.6	19.7	23.1	26.4	27.0	27.9	23.7	19.2	13.5	5.7		
	10%	WB	2.2	4.8	7.2	12.5	16.6	19.5	20.0	19.6	17.0	13.3	9.2	3.8		
		MCDB	2.8	6.2	10.1	17.5	21.8	24.9	25.6	26.3	22.4	16.6	11.1	4.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.7	8.4	10.0	11.5	11.6	11.7	12.0	12.7	11.5	10.9	8.1	5.7		
		MCDBR	8.0	11.1	14.1	15.7	14.8	14.3	14.6	15.7	15.5	15.4	12.2	7.2		
		MCWBR	6.0	7.3	8.1	7.3	6.9	6.3	5.9	5.5	5.9	8.6	7.4	5.6		
		MCWBR	6.9	10.2	12.7	13.9	13.4	13.2	13.2	14.0	13.6	13.7	11.1	6.7		
(40) Clear-Sky Solar Irradiance	taub	taub	0.306	0.335	0.378	0.430	0.414	0.410	0.431	0.434	0.394	0.361	0.335	0.308		
		taud	2.312	2.256	2.219	2.135	2.251	2.293	2.240	2.230	2.306	2.382	2.393	2.362		
		Ebn at Noon	768	818	834	821	852	856	830	810	810	783	730	721		
		Edh at Noon	75	98	120	143	132	128	133	129	109	87	70	65		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.23	2.06	3.16	4.22	5.19	5.55	5.52	5.09	3.64	2.48	1.40	0.94		
		RadStd	0.18	0.23	0.40	0.42	0.51	0.46	0.50	0.47	0.45	0.33	0.19	0.13		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (60 neighbors)	+0.74	N/A	N/A	+1.03	N/A	N/A	-156	-226	+113	+40

Nomenclature: See separate page